

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions to drive smarter retail decisions.

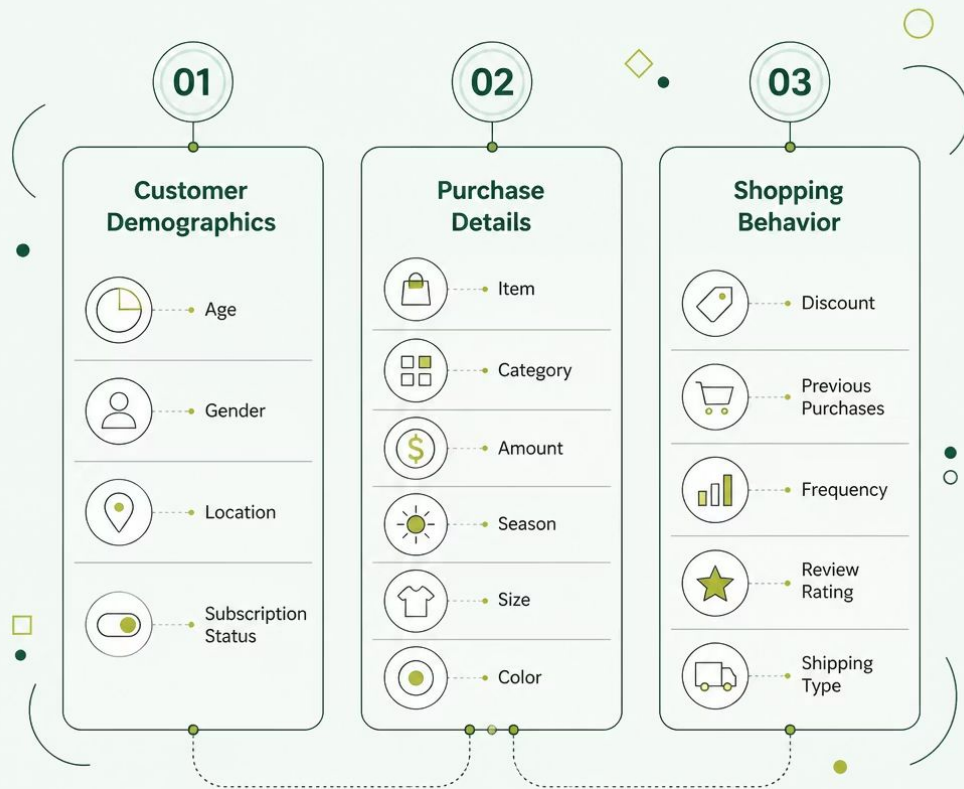
DATA ANALYTICS

PYTHON · SQL · POWER BI



Made with GAMMA

Dataset at a Glance



3,900 Transactions

Across 18 columns covering demographics, purchase details, and behavioral signals.

- **25 unique items** across 4 categories
- **50 locations** represented
- **37 missing values** in Review Rating — imputed by category median
- Average purchase: **\$59.76**
- Average rating: **3.75 / 5**

Data Preparation in Python

1

Load & Explore

Imported via `pandas`; inspected structure with `df.info()` and `.describe()`

2

Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to `snake_case`

3

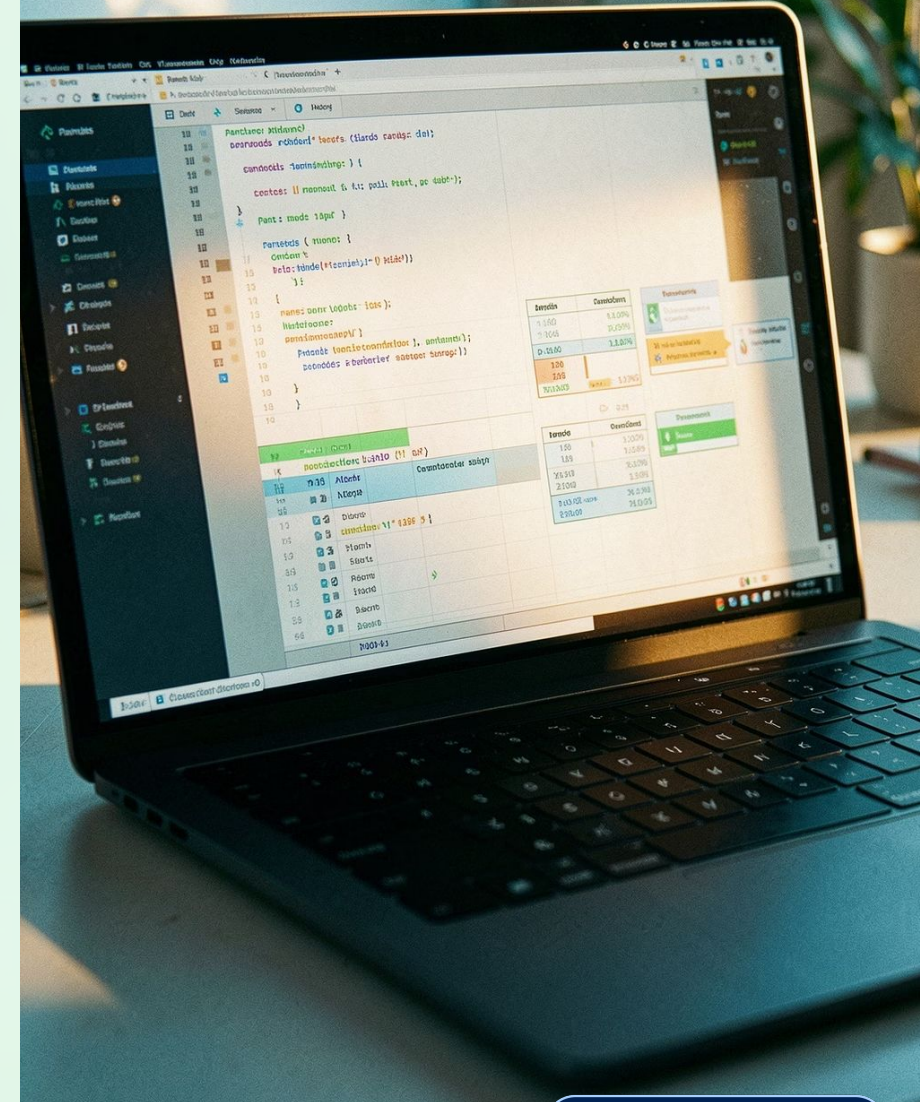
Engineer Features

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`

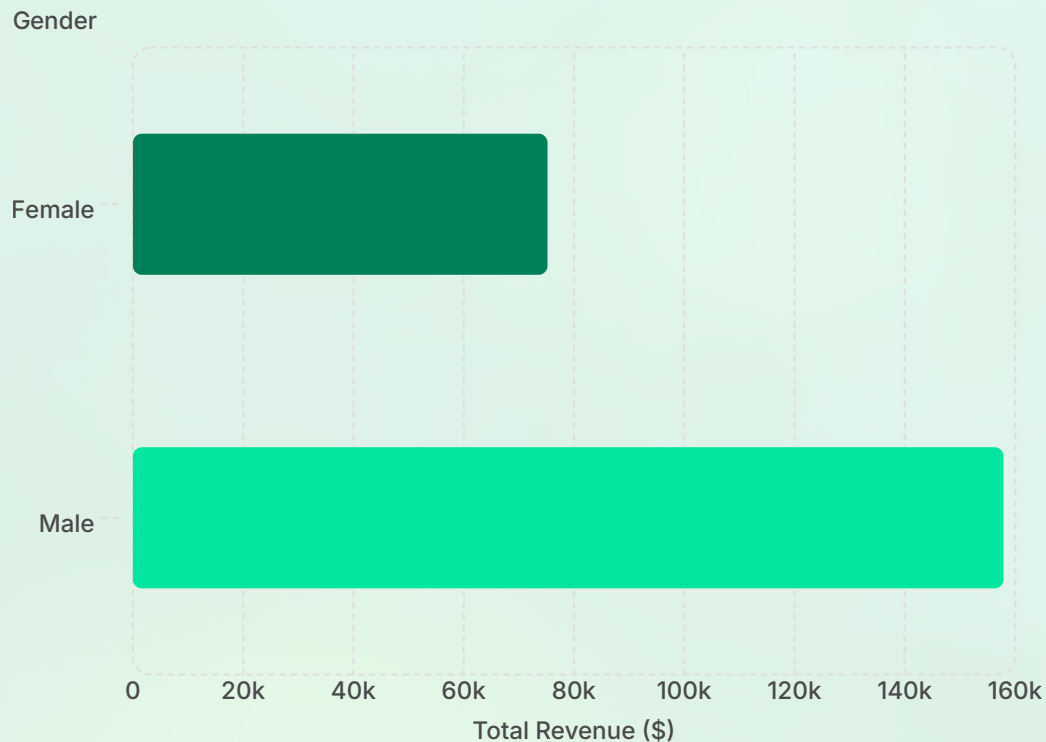
4

Load to PostgreSQL

Cleaned DataFrame loaded into PostgreSQL for structured SQL analysis



Revenue & Spending Insights



Male Customers Drive 2× Revenue

Male customers generated **\$157,890** vs. **\$75,191** from female customers — a significant gap worth investigating for targeted campaigns.

High-Spending Discount Users

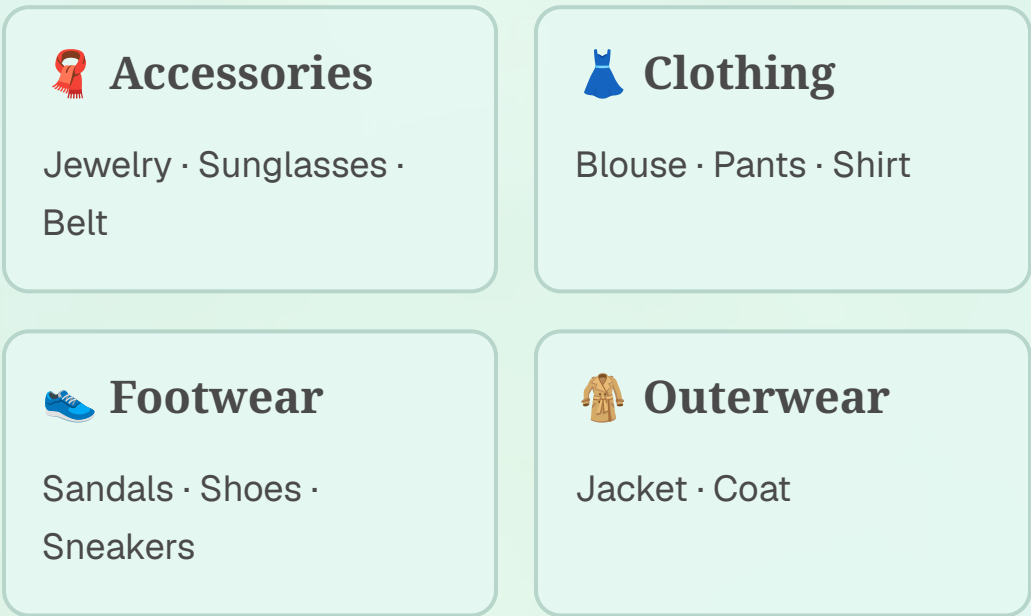
839 customers used discounts yet spent above the average purchase amount — a segment primed for upsell offers without additional incentives.

Product Performance

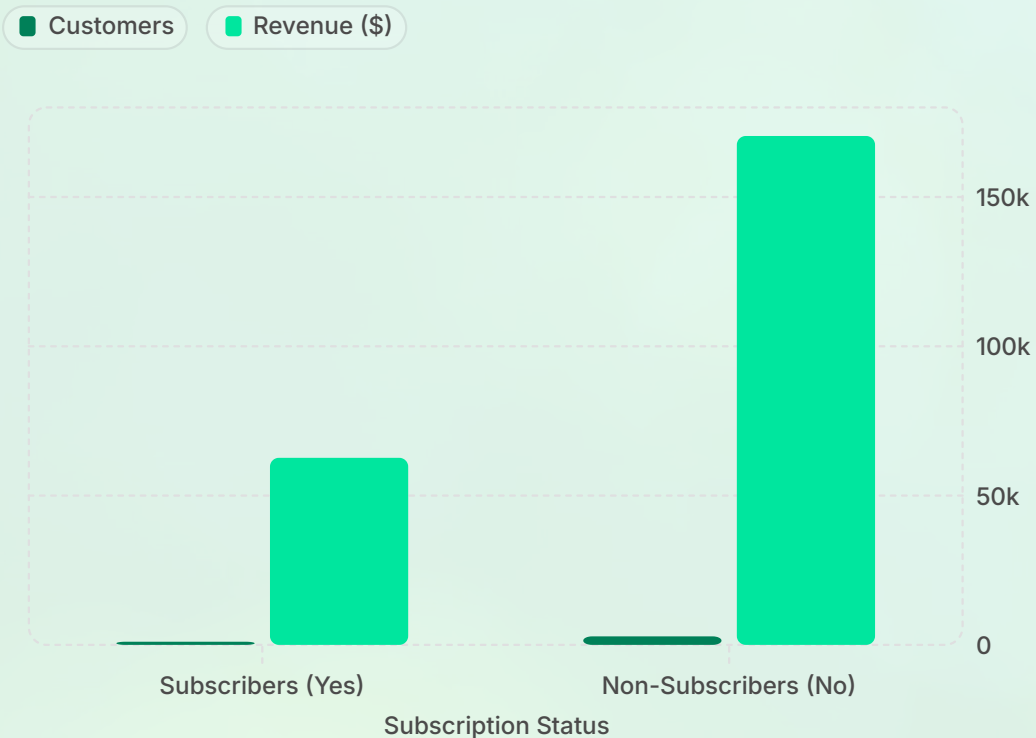
Top 5 Products by Average Rating



Top 3 Products per Category



Subscription & Discount Behavior



Subscribers Spend Slightly Less Per Transaction

Average spend is nearly identical — **\$59.49** (subscribers) vs. **\$59.87** (non-subscribers) — but non-subscribers dominate volume.

Most Discount-Dependent Products

- Hat** — 50% of purchases discounted
- Sneakers** — 49.66% discounted
- Coat** — 49.07% discounted

Customer Segmentation

3,116

Loyal Customers

80% of the base — strong retention
foundation

701

Returning

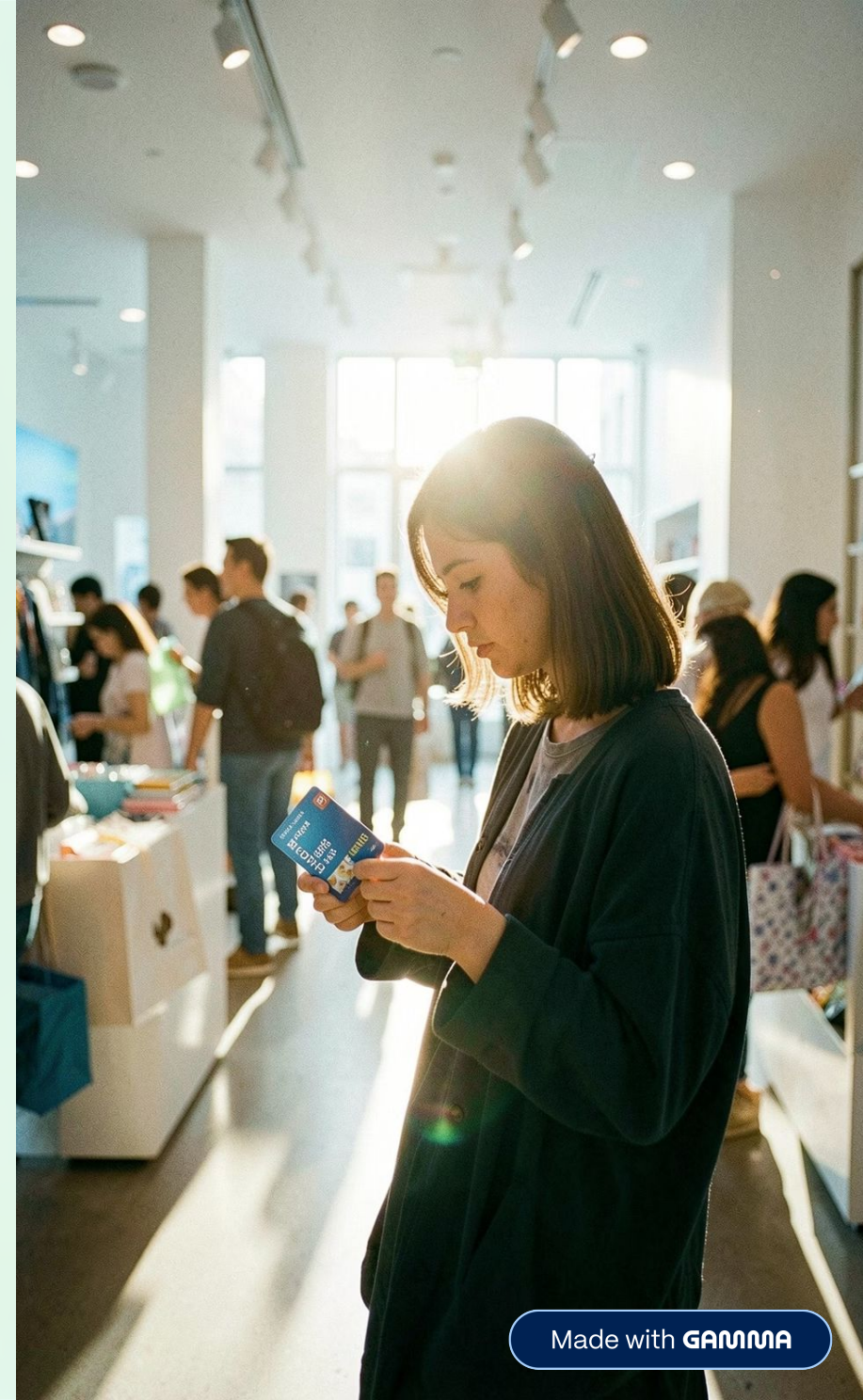
18% — prime candidates for loyalty
upgrades

83

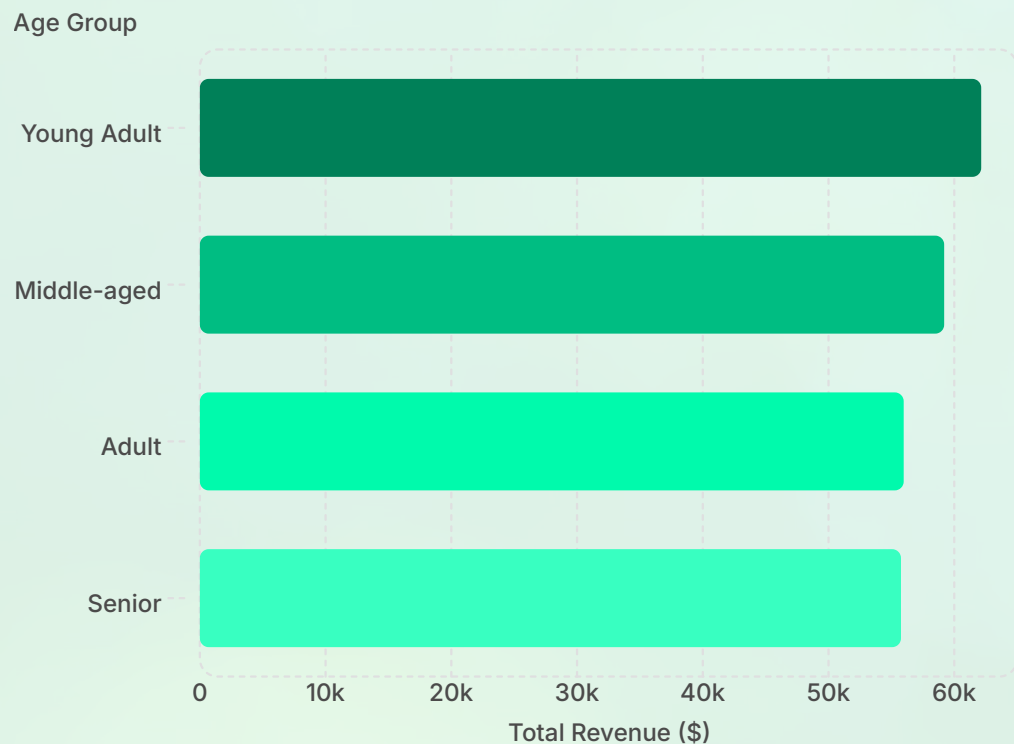
New Customers

2% — onboarding and first-purchase
incentives critical

📌 **Key finding:** Of customers with 5+ purchases, 958 are subscribers vs. 2,518 non-subscribers — repeat buyers are not yet converting to subscriptions at scale.



Revenue by Age Group & Shipping



Young Adults Lead Revenue

Young Adults contribute the highest revenue at **\$62,143**, but all four age groups are relatively balanced — suggesting broad market appeal.

Express Shipping Commands a Premium

Average order value for Express shipping (**\$60.48**) edges out Standard (**\$58.46**), indicating urgency-driven buyers spend more.

Power BI Dashboard



Interactive Customer Behavior Dashboard

Built in Power BI to surface key metrics at a glance with drill-down filters.

Filters include Subscription Status, Gender, Category, and Shipping Type for segmented analysis.

3.9K

Total Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Business Recommendations

Boost Subscriptions

Promote exclusive perks to convert the 2,518 repeat non-subscribers.

Loyalty Programs

Reward Returning customers (701) to accelerate movement into the Loyal segment.

Review Discount Policy

50% of Hat purchases use discounts — evaluate margin impact and explore targeted alternatives.

Targeted Marketing

Prioritize Young Adults and Express shipping users for high-value campaigns.

Product Positioning

Feature top-rated items (Gloves, Sandals, Boots) prominently in campaigns and recommendations.